Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 403
Status Update

Inverted Pendulum Robot 1
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Gage Fox, Eric Chang
TA: Ay Fahrettin
Sponser: Aniruddha Datta

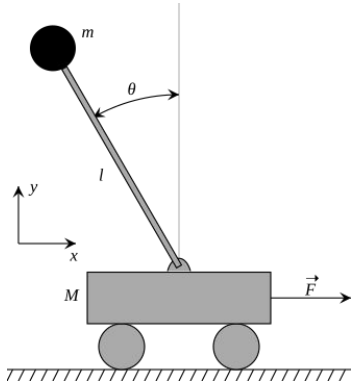
Project Summary

Problem statement:

- Autonomous warehouse robots generally experience instability when handling unbalanced loads or making sudden maneuvers. These movements can cause tilting, making balance a key challenge.

Our Solution Will:

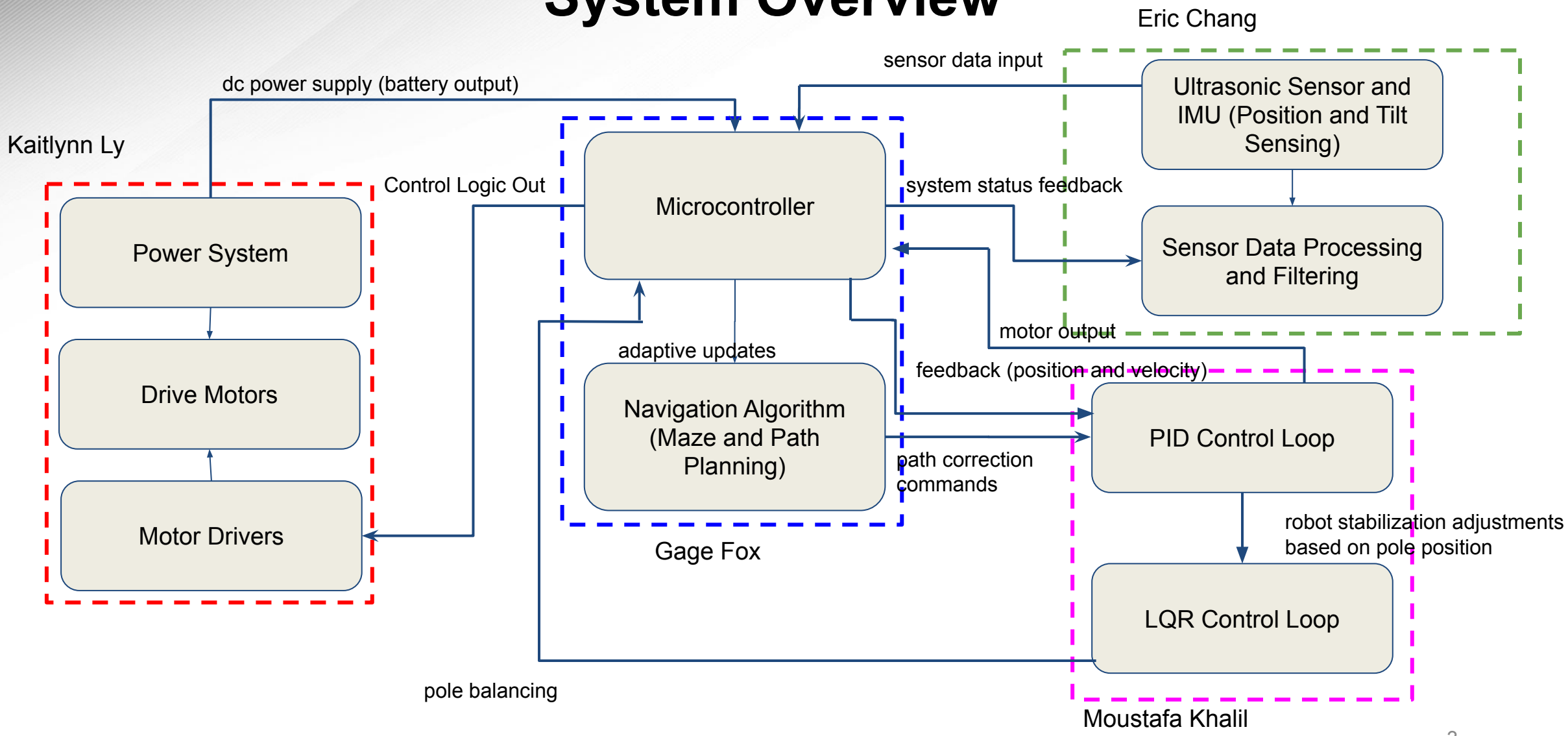
- Deliver an autonomous, self-navigating robot that maintains stability of an unstable payload while moving through constrained spaces.
- Integrate mechanical, sensing, power, and control subsystems into a cohesive, microcontroller-based platform designed for reliability and safety.



Side View

Isometric View

System Overview



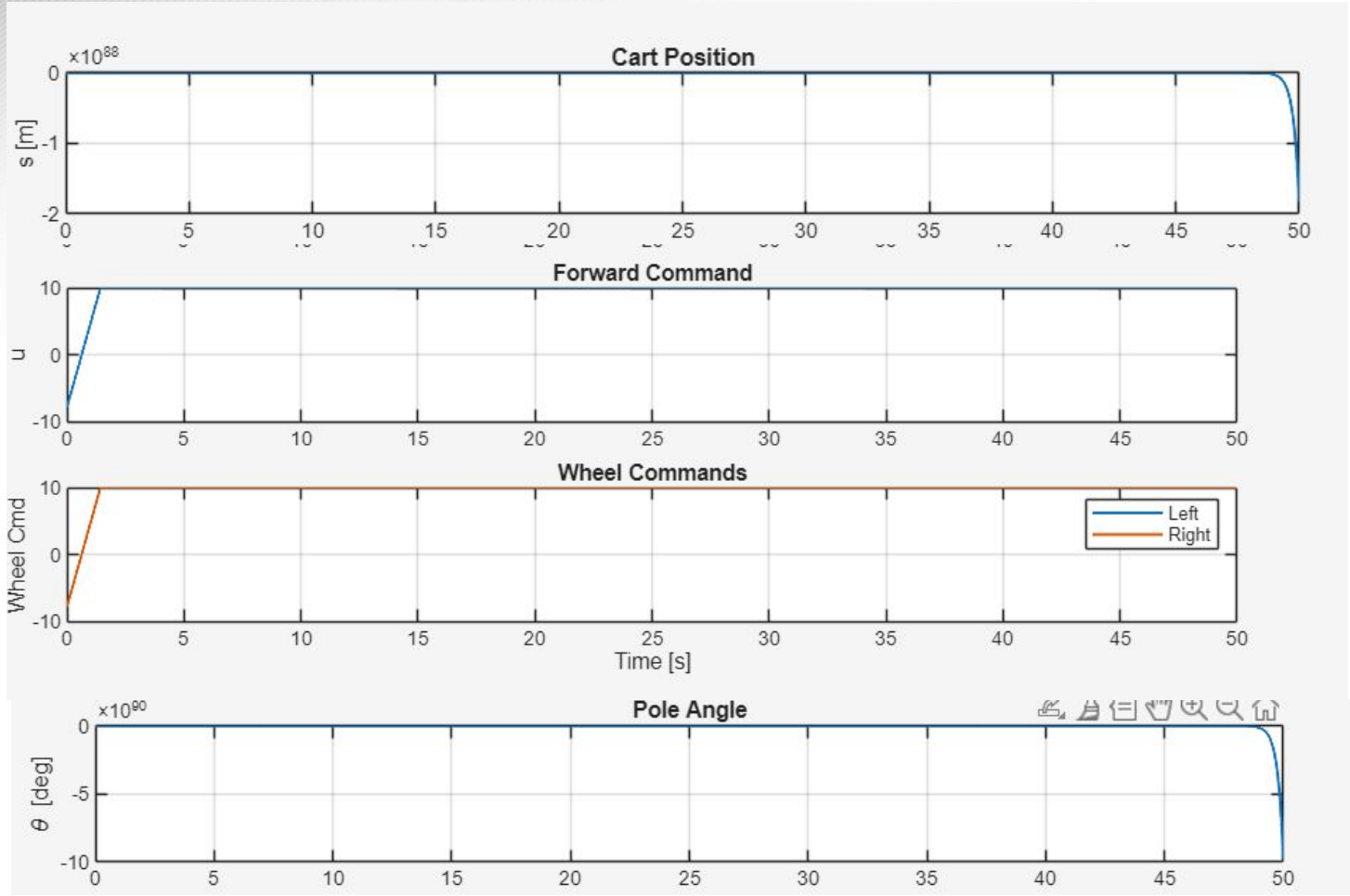
Controls Subsystem

Moustafa Khalil

Accomplishments since midterm update	Ongoing progress/problems and plans until the next presentation
• Created the Control Algorithm • Coded and simulated the system in MATLAB, generating time response graphs • Analyzed system stability and tuned controller parameters through iterative simulation	• Create Simulink representation of the MatLab code • Finalize control tuning, validate stability across multiple operating conditions

Controls Subsystem

Moustafa Khalil



Pole Angle x Time

Forward Command x Time

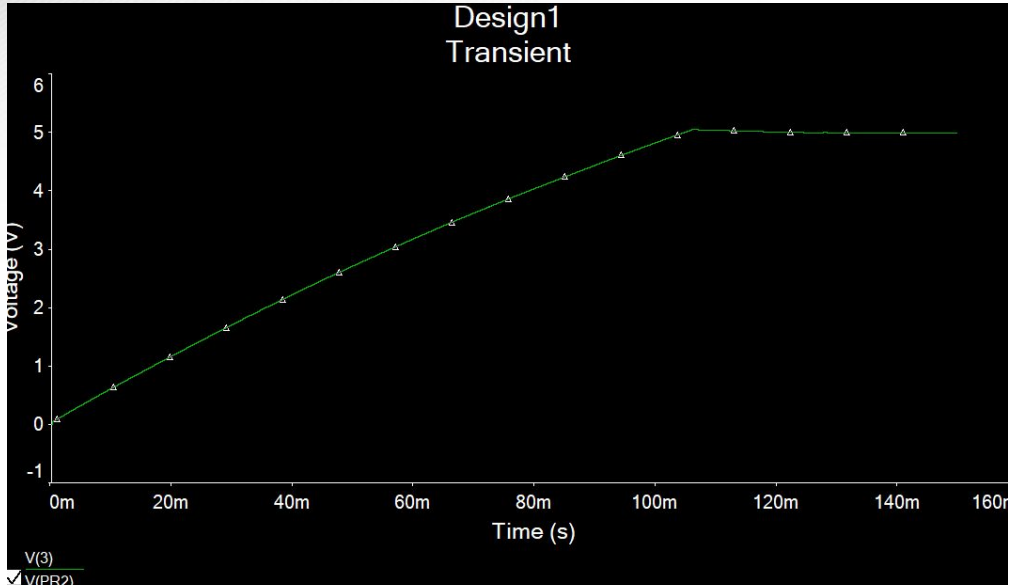
Power & Mechanical

Kaitlynn Ly

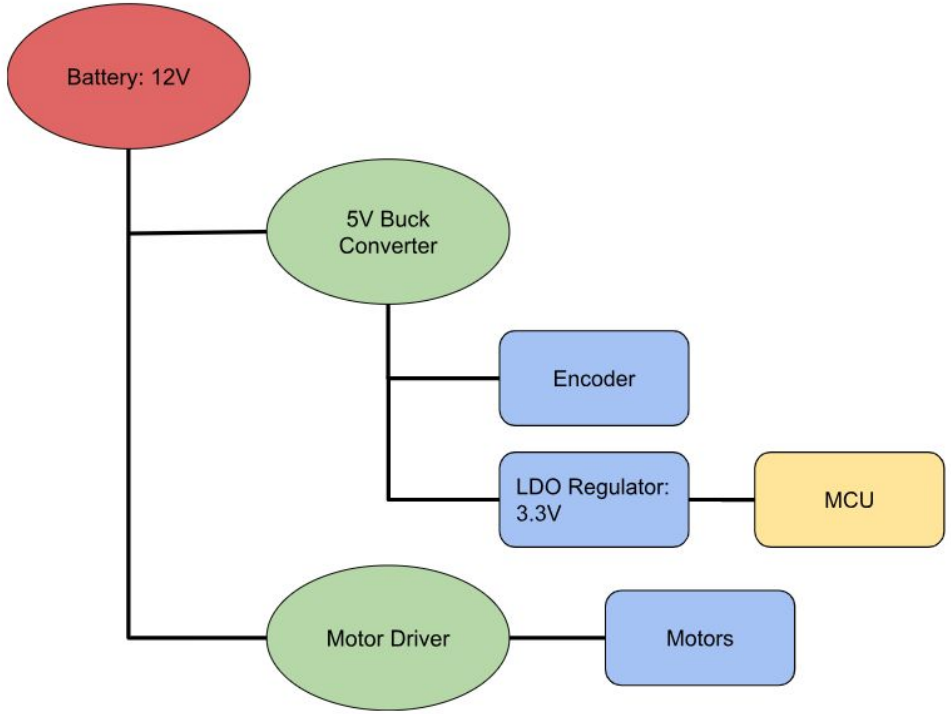
Accomplishments since midterm update	Ongoing progress/problems and plans until the next presentation
• Buck converter design and simulation with integrated LC filter • Power distribution architecture for full robot system • Robot layout and structure	• Linear dropout regulators to step down 5V to 3.3V • Testing motor driver operation and motor behavior • Conduct power integrity and efficiency testing

Power & Mechanical

Kaitlynn Ly



Buck converter output voltage over time



Power distribution architecture

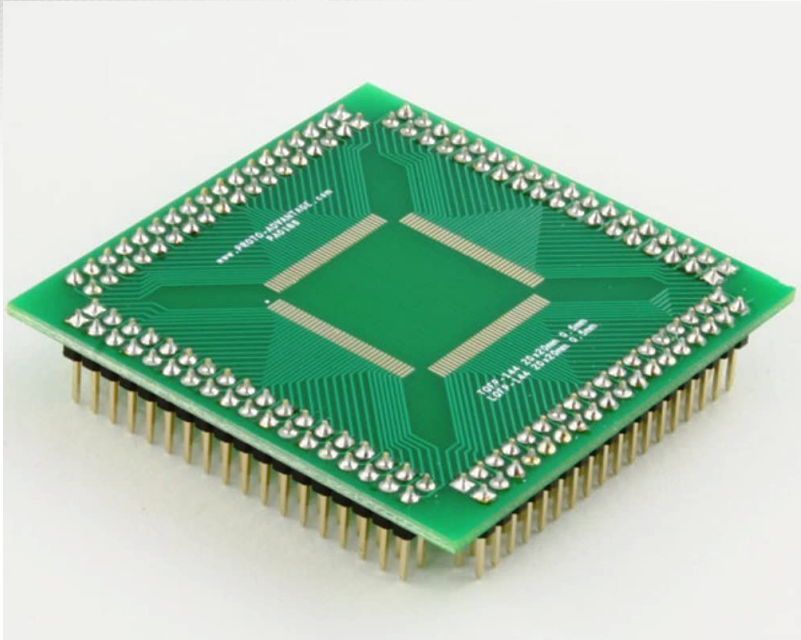
Microcontroller

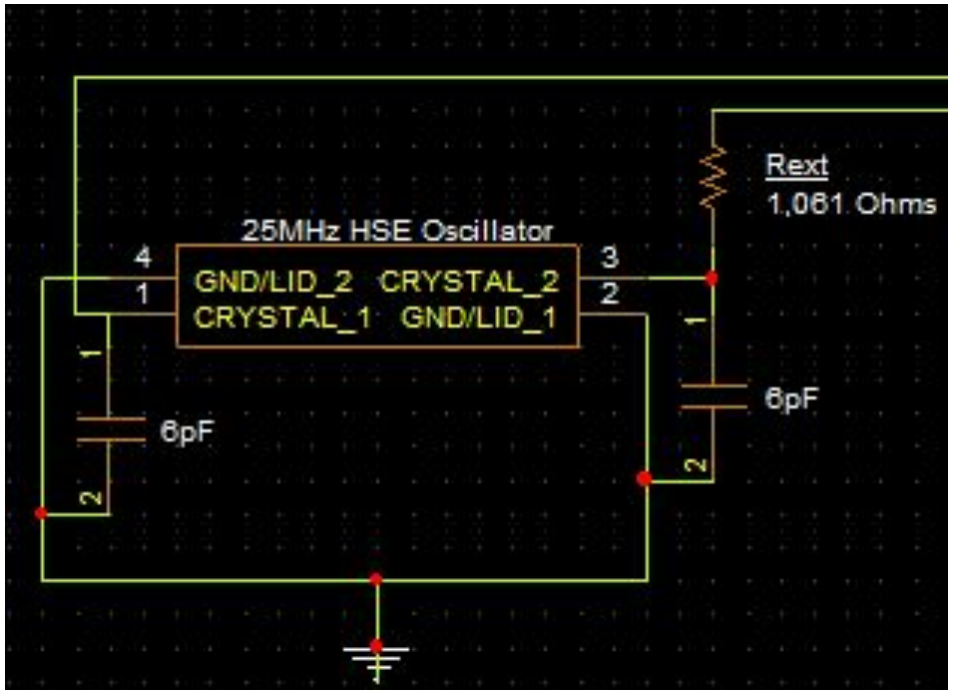
Gage Fox

Accomplishments since midterm update	Ongoing progress/problems and plans until the next presentation
• Ordered all major components for MCU interfacing and programming. (All parts received). • Determined necessary capacitors and resistor for HSE crystal resonator interfacing. • Determined necessary communication protocols and wiring from peripheral sensors to MCU.	• Currently having Dr. Lusher solder MCU to LQFP144-PGA144 breakout. • Working on PCB layout and traces in OrCAD Capture. • Considering using different crystal oscillator that is more stable. • Not currently able to test anything due to prototyping parts coming in late.

Microcontroller

Gage Fox





- STM32F407ZGT7 has many interfacing issues due to it being a standalone chip

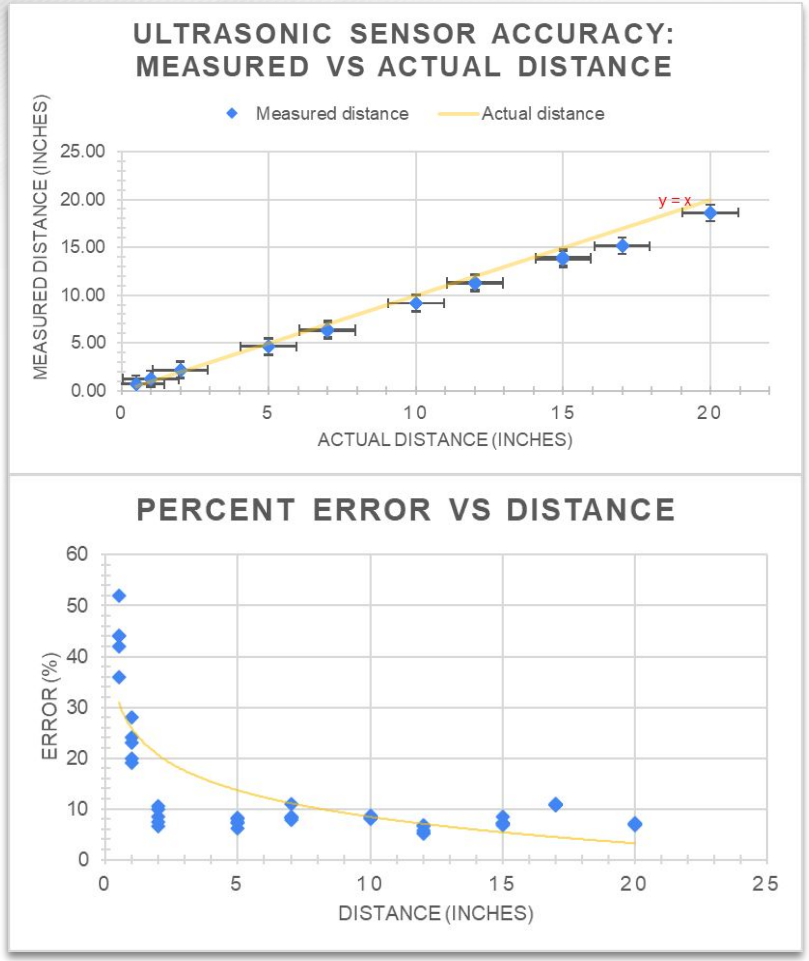
- The current oscillator design could be unstable

Sensors and Navigation

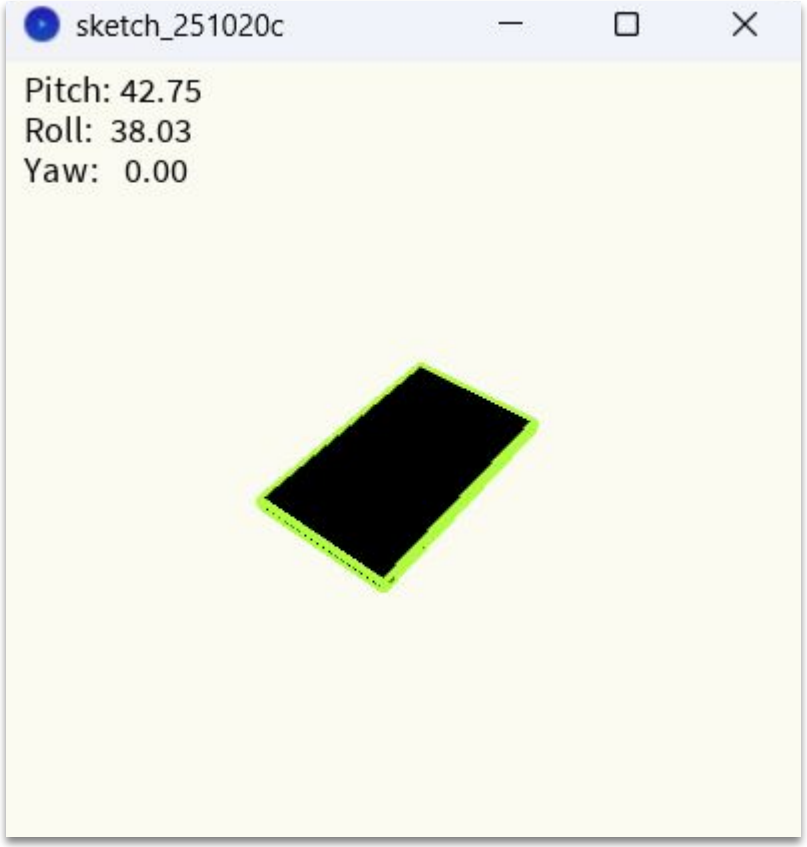
Eric Chang

Accomplishments since midterm update	Ongoing progress/problems and plans until the next presentation
• Interfaced ultrasonic sensor and Inertial Measurement Unit (IMU); stored in CSV format for analysis • Conducted error analysis on ultrasonic sensor data; validating ≥95% obstacle detection at distances ≥2 cm • Designed basic Extended Kalman Filter (EKF) for sensor fusion using dummy data	• Implementation of code into ROS for real-time analysis • Tuning and calibration of raw sensor data with moving average filter and EKF • Utilize Navigation Toolbox in MATLAB to fuse IMU and sensors for path planning and localization

Sensors and Navigation



Ultrasonic sensor performance and accuracy analysis across varying distances



Real-time IMU visualization and sensor data collection

Execution plan

3	Subsystem Development and Testing					
3.1	Power and mechanical subsystem construction	Kaitlynn L.				
3.1.1	Design buck converter circuit	Kaitlynn L.	10/6/25	10/10/25	4	80%
3.1.2	Assemble, solder, and test buck converter	Kaitlynn L.	10/19/25	10/20/25	1	20%
3.1.3	Cut and join aluminum and plywood chassis	Kaitlynn L.	10/19/25	10/20/25	1	0%
3.1.4	Mount and install power electronics	Kaitlynn L.	10/27/25	11/3/25	6	0%
3.1.5	Mount and test pendulum	Kaitlynn L.	10/27/25	11/3/25	6	0%
3.1.6	Subsystem validation	Kaitlynn L.	11/03/25	11/08/25	5	0%
3.2	Controls subsystem construction	Moustafa K.				
3.2.1	Create LQR and PID control algorithm based of control model	Moustafa K.	10/06/25	10/15/25	9	100%
3.2.2	Code the control algorithm	Moustafa K.	10/15/25	10/27/25	12	80%
3.2.3	Subsystem validation	Moustafa K.	10/27/25	11/08/25	12	0%
3.3	Sensors & navigation subsystem contruction	Eric C.				
3.3.1	Setup sensor interfaces (I^2C communication)	Eric C.	10/06/25	10/12/25	6	100%
3.3.2	Test raw data; design filtering and fusion algorithms	Eric C.	10/13/25	10/26/25	6	70%
3.3.3	Test filtering algorithms with MATLAB	Eric C.	11/03/25	11/09/25	6	10%
3.3.4	Validate sensors, filters, GUI; compare with metrics	Eric C.	11/17/25	11/23/25	6	0%
3.4	MCU subsystem construction	Gage F.				
3.4.1	MCU circuit design	Gage F.	10/06/25	10/15/25	9	100%
3.4.2	MCU interfacing, and protocols	Gage F.	10/06/25	10/15/25	9	60%
3.4.3	PCB Design	Gage F.	10/27/25	11/03/25	7	0%
3.4.4	Blutooth implementation for MCU	Gage F.	10/27/25	11/03/25	7	0%
3.4.5	Subsystem Validation	Gage F.	11/03/25	11/08/25	5	0%

Validation plan

Task	Specifications and Success Criteria	Status	Responsible Engineer(s)
Logic Calculation Rate	Validate through timing tests on MCU execution cycle; must complete calculations in < 50ms	Untested	Gage
Data Transfer Rates	Measure message latency with time logs; must be < 100 ms difference	Untested	Gage
Oscillator Precision	Validate clock stability through frequency tests; ensure deviation < 10ppm	Untested	Gage
Sonar Read Distance	Validate using logged ultrasonic data (CSV) and IMU; confirm ≥95% obstacle detection at distances ≥2 cm and ±3 cm accuracy within 0.2-1.0 m	Tested	Eric
Motor Precision	Left/right wheel speed mismatch ≤ 3% over 2 m straight; speed tracking error ≤ ±5%	Untested	Eric
Verify Input Power Delivery	Measure system voltage under load; confirm range is within 11.1-12.6V	Untested	Kaitlynn
Buck Converter Regulation	Record output voltage under different loads; verify 5V stability	Untested	Kaitlynn
LDO Validation	Measure regulated output with oscilloscope; confirm 3.3V stability	Untested	Kaitlynn
Motor Current Capacity	Monitor motor current draw during operating mode; ensure <7A	Untested	Kaitlynn
Control Algorithm Validation	Validate stability across various operating conditions and confirm tuned gains achieve target response time	Untested	Moustafa

Thank You!

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Any Questions?